

Week 4: Mini-Project — Algorithm Challenge Set

Goal

Solve 5 algorithm challenges. But the code is only half the job — you must also **document your thinking process** using the UPER framework for each problem.

This is what separates someone who can code from someone who can **solve problems**.

Setup

Start with these variables:

```
let numbers = [12, -5, 8, 20, 3];  
let text = "bootcamp";  
let tempCelsius = 30;
```

The 5 Challenges

For each challenge, write 4 sections:

1. **Understand** — What are the inputs? What should the output be? Any edge cases?
 2. **Plan** — Write pseudocode (no JavaScript yet)
 3. **Execute** — Write the JavaScript code
 4. **Reflect** — What could go wrong? How could you improve it?
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Challenge 1: Find the Smallest Number

Find and print the smallest number in the `numbers` array.

Expected output: `-5`

Solution with UPER

****Understand:****

- Input: Array of numbers `[12, -5, 8, 20, 3]`
- Output: The smallest number (`-5`)
- Edge case: What if the array was empty?

****Plan:****

...

```
SET smallest TO first element of array
FOR EACH number IN array
  IF number < smallest THEN
    SET smallest TO number
  ENDIF
ENDFOR
PRINT smallest
...
```

****Execute:****

```
```javascript
let smallest = numbers[0];
for (let i = 0; i < numbers.length; i++) {
 if (numbers[i] < smallest) {
 smallest = numbers[i];
 }
}
console.log("Smallest: " + smallest); // -5
```
```

****Reflect:**** We start `smallest` at `numbers[0]`, not at 0. If we started at 0 and all numbers were positive (like `[5, 10, 15]`), the answer would wrongly be 0 — a number that's not even in the array.

Challenge 2: Calculate the Average

Calculate and print the average of all numbers in the array.

Expected output: 7.6

Solution with UPER

****Understand:****

- Input: Array of numbers
- Output: Average (sum / count)
- Edge case: Empty array would cause division by zero

****Plan:****

...

```
SET total TO 0
FOR EACH number IN array
```

```
ADD number TO total
ENDFOR
SET average TO total / length of array
PRINT average
...
```

```
**Execute:**
```javascript
let total = 0;
for (let i = 0; i < numbers.length; i++) {
 total = total + numbers[i];
}
let average = total / numbers.length;
console.log("Average: " + average); // 7.6
...
```

**\*\*Reflect:\*\*** If the array were empty (`numbers.length === 0`), we'd divide by zero, giving `NaN`. In a real app, you'd check for that first.

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### Challenge 3: Search for a Number

Check if the number 8 exists in the array. Print `true` or `false`.

**Expected output:** `true`

#### *Solution with UPER*

**\*\*Understand:\*\***

- Input: Array of numbers + target number (8)
- Output: `true` if found, `false` if not
- Edge case: Target appears multiple times (still just true)

**\*\*Plan:\*\***

```
...
SET target TO 8
SET found TO FALSE
FOR EACH number IN array
 IF number EQUALS target THEN
 SET found TO TRUE
 STOP LOOP
ENDIF
ENDFOR
PRINT found
...
```

```
Execute:
```javascript
```

```
let target = 8;
let found = false;
for (let i = 0; i < numbers.length; i++) {
  if (numbers[i] === target) {
    found = true;
    break;
  }
}
console.log("Found: " + found); // true
...

```

****Reflect:**** The `break` is important for performance. If the array had 1 million numbers and 8 was at position 3, we'd stop after 3 checks instead of looping through all 1 million.

Challenge 4: Remove Vowels

Remove all vowels from `text` and print the result.

Expected output: `"btcmp"`

Solution with UPER

****Understand:****

- Input: String "bootcamp"
- Output: String with vowels removed ("btcmp")
- Edge case: Uppercase vowels? We should handle both.

****Plan:****

```
...
SET result TO ""
FOR EACH letter IN text
  IF letter is NOT a vowel THEN
    ADD letter TO result
  ENDIF
ENDFOR
PRINT result
...

```

****Execute:****

```
```javascript
let result = "";
for (let i = 0; i < text.length; i++) {
 let letter = text[i].toLowerCase();
 if (letter !== 'a' && letter !== 'e' && letter !== 'i' && letter !== 'o' && letter !== 'u') {
 result = result + text[i];
 }
}
}

```

```
console.log("Without vowels: " + result); // "btcmp"
````
```

****Reflect:**** We use `&&` (AND), not `||` (OR). With OR, the condition "is not a AND is not e" would be true for every letter — because 'a' is not 'e', and 'e' is not 'a'. Every vowel would pass the check. This is a common logic trap.

Challenge 5: Temperature Converter

Convert `tempCelsius` to Fahrenheit. Formula: $(C * 9/5) + 32$.

Expected output: `"30C = 86F"`

Solution with UPER

****Understand:****

- Input: Temperature in Celsius (30)
- Output: Temperature in Fahrenheit (86)
- This is a straight calculation — no loops or conditions needed

****Plan:****

```
SET fahrenheit TO (celsius * 9/5) + 32
PRINT celsius + "C = " + fahrenheit + "F"
````
```

**\*\*Execute:\*\***

```
````javascript
let fahrenheit = (tempCelsius * (9 / 5)) + 32;
console.log(tempCelsius + "C = " + fahrenheit + "F"); // "30C = 86F"
````
```

**\*\*Reflect:\*\*** This is the simplest type of algorithm — purely sequential, no branches or loops. Not every problem needs complex logic.

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## How to Submit

1. Write all 5 solutions in a single file
2. Include your UPER documentation as comments above each solution
3. Take a screenshot showing all 5 outputs in your console
4. Upload to Google Classroom

**Example format:**

```
// === Challenge 1: Find Smallest ===
// Understand: Input is array of numbers, output is the smallest one
// Plan: Start with first element, loop through, update if smaller found
// (your code here)
// Reflect: Must start at numbers[0], not 0

// === Challenge 2: Average ===
// ...
```